

Name: _____

Homework 1

Show your work. Include any code snippets that you used to generate answers. Complete this assignment individually.

1) What are the two main types of attributes typically found in data.
[2 points]

2) Consider the following matrix D ; and answer all the questions. [14 points]

	X_1	X_2	X_3
x_1	1.2	3	1.7
x_2	2.4	5	2.4
x_3	4.8	35	1.2
x_4	6.6	60	3.1
x_5	-0.5	24	3.3
x_6	3.4	32	8.4
x_7	2.1	1	6.5

a. What is the estimated mean of this data attribute X_1 ? [2 points]

b. What is the estimated covariance between attributes X_2 and X_3 ? [2 points]

c. What is the estimated multi-dimensional mean $\hat{\mu}$ of this data matrix? (Your answer should be a vector) [2 points]

d. What is the estimated covariance $\hat{\sigma}_{12}$ of the attributes X_1 and X_2 ? [2 points]

e. What is the covariance matrix of D ? [2 points]

f. What is the correlation coefficient between the attributes X_1 and X_2 ? [2 points]

g. What is the total variance of this matrix D ? [2 points]

3) Consider the following 5-dimensional vectors: [6 points]

$$a = (1.2 \quad -2.3 \quad 4 \quad 7.1 \quad -3.12)$$

$$b = (23.2 \quad 3 \quad 1.2 \quad -3.21 \quad 5)$$

$$c = (8.2 \quad -4.6 \quad 2 \quad 1 \quad -2)$$

a. What is the $\|a - c\|_2$? [2 points]

b. What is the $\|b - a\|_1$? [2 points]

c. What is the angle between the vectors a and c ? [2 points]

4) Consider the following matrix D.

$$D = \begin{pmatrix} & X_1 & X_2 & X_3 \\ x_1 & 36.6 & Mild & 32 \\ x_2 & 38 & Severe & 21 \\ x_3 & 37 & Extreme & 67 \\ x_4 & 39 & Extreme & 11 \\ x_5 & 27 & Moderate & 71 \end{pmatrix}$$

- a. Use the One-Hot encoding method to transform the categorical data into numerical data in the following matrix. You can assume that the attribute X_2 can only contain 4 values: $\{Mild, Moderate, Severe, Extreme\}$ [2 points]

- b. What is the value of $\|x_3 - x_5\|$ (based on the transformed data matrix)? [2 points]

5) The following questions reference the Heart Disease data set from the UCI Machine Learning Repository:

<https://archive.ics.uci.edu/ml/datasets/Heart+Disease>

Answer the following questions about the dataset.

- a. How many rows (entities/instances) are there in this dataset? [1 point]

- b. How many attributes are in this dataset? [1 point]

- c. What is kind of data is stored in the attribute “cigs”? [1 point]